



DPUCANMSG

Allows bitwise definition of an arbitrary CAN message. The message can be sent or received periodically.

1. In- /Outputs:

Value[1...32]

Input or output value of a field.

DATA[1...8]

Raw data (bytes) of the CAN bus packet.

2. Dynamic parameters:

ValueMultiplier[1...32]

Multiplier for the field.

ValueOffset[1...32]

Offset for the field.

DataBegin[1...32]

Start position of the field within the CAN bus packet, in bits.

DataLength[1...32]

Length, in bits, of the field.

3. Static parameters:

CANMode

CAN bus channel over which the message is sent or received (see DPUCANBUS).

ID

ID of the message.

LEN

Length of the message (in bytes).

Write

Defines if the message is written to the bus (1 or 2, see below) or read from the bus (0).

Period

If Write=1: Period (in ms), in which the data is written.

SendOnce

If Write=2: Whenever this signal rises from 0 to 1, the message is sent once.

4. Dependencies to other DPUs:

In addition to DPUCANDBC, a DPUCANBUS with the same CANMode must be running on the same computer, which is connected to the CAN hardware. The *Index* value of DPUCANBUS must be lower than the Index value of DPUCANDBC.

5. Functional principle:

The DPU composes the CAN bus message from at most 32 data fields (*ValueXX*), which are inserted in the up to 8 data bytes of the message.

The start bit of each field is set using the parameter *DataBeginXX*, its length is set using *DataLengthXX*. The data is interpreted as integers and transformed using the multiplier and the offset parameters, as follows:



If the message is read from the CAN bus:

$$\text{Value} = (\text{Integer}(\text{DATA}, \text{Begin}, \text{Length}) * \text{ValueMultiplier}) + \text{ValueOffset}$$

If the message is written to the CAN bus:

$$\text{Integer}(\text{DATA}, \text{Begin}, \text{Length}) = (\text{Value} - \text{ValueOffset}) / \text{ValueMultiplier}$$

6. Example:

```
DPUCANBUS ~localCAN {  
    Computer = { LOCALHOST };  
    Index = 11;  
  
    CANMode = 0;  
    CANType = 0; # standard  
    Baud = 1; # 500 kBaud  
};
```

```
DPUCANMSG MSG_549 {  
    Computer = { LOCALHOST };  
    Index = 12;  
  
    CANMode = 0;  
    ID = 549;  
    LEN = 8;  
    Write = 1;  
    Period = 100;  
};
```